

Limited Degradation of Active Pharmaceutical Ingredient After Spaceflight on the International Space Station

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Abstract

Introduction: Spaceflight may accelerate degradation of active pharmaceutical ingredients, raising concerns about medication stability during long-duration missions. This study assessed the potency of active pharmaceutical ingredients in a convenience sample of medications flown to the International Space Station alongside lot-matched terrestrial control samples.

Methods: Spaceflight-exposed samples and terrestrial control samples were analyzed using ultra-high-performance liquid chromatography with tandem mass spectrometry. At the time of analysis, all medications were beyond their labeled expiration dates following prolonged storage and delayed laboratory testing. Medications included caffeine, diazepam, diphenhydramine, epinephrine, ketamine, lidocaine, naloxone, and promethazine in solid and liquid formulations. Percent difference between exposed and control samples was calculated, and absolute potency was evaluated relative to 95 to 105% and 80 to 120% ranges. Regression analyses examined associations of changes in the potency of active pharmaceutical ingredients with days in space and days expired.

Results: Most samples from both groups fell outside the 95 to 105% range. However, >90% remained within the broader 80 to 120% range. Average percent differences between spaceflight-exposed and ground control samples were generally within $\pm 5\%$ for most medications, suggesting limited relative potency differences despite prolonged spaceflight exposure. Epinephrine exhibited the most consistent reductions, averaging 10% lower potency than control samples. Regression analyses identified medication-specific associations with days expired and days in space but did not show a consistent pattern of spaceflight-associated potency loss.

Conclusion: Within the limitations of this lot-matched, post-storage comparison, International Space Station exposure was not associated with large relative potency changes for most medications studied. Epinephrine represented the clearest exception and may require medication-specific stabilization strategies for long-duration space exploration missions.

Keywords

medication, pharmaceutical, stability, spaceflight, aerospace medicine

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Introduction

The Health Maintenance System (HMS) on the International Space Station (ISS) represents the most robust low-Earth-orbit medical capability in use to date.^{1,2} The HMS contains a selection of up to 190 pharmaceuticals that are regularly used, with >500 doses of at least 30 pharmaceuticals deployed across 33 ISS missions.^{3,4} Providing medical care in space faces unique challenges compared with other analogues and austere locations. Although the ISS is a carefully controlled spacecraft environment, medications in the HMS are exposed to high ambient carbon dioxide levels, continuous solar and galactic cosmic radiation exposures, and microgravity.⁵ Additionally, given mass and volume restrictions at launch, many solid oral medications (eg, tablets and capsules) are removed from sealed manufacturer packaging and placed into nonprotective alternate packaging prior to flight, often as simple as zipper-locked plastic bags.^{6,7}

Previous research has raised concerns that the unique conditions of the spaceflight environment may lead to accelerated degradation of active pharmaceutical ingredients (API).^{8–13} One notable study showed that 25 of 36 medications exposed to up to 880 d of spaceflight aboard the ISS fell below Food and Drug Administration reference standards for labeled API strength.⁸ To date, however, the rate of degradation and the specific mechanisms by which spaceflight may facilitate loss of API potency remain largely uncharacterized.⁵ Potential spaceflight-related medication degradation is mitigated operationally on the ISS by routine resupply, which helps maintain in-date pharmaceuticals for clinical use within the HMS.^{1,14} This resupply process provided an opportunity to examine operationally deployed medications in a variety of drug delivery forms (ie, tablets, capsules, and liquids), repackaged to NASA flight standards, along with lot-matched ground control samples, across multiple ISS missions. Because these medications were retained in operational use aboard the ISS, many were already at or near the end of their labeled shelf life at the time of return to Earth. An unplanned pandemic-related interruption further delayed final analytic testing. We report the results of an exploratory analysis of relative changes in API content in a convenience sample of operationally deployed ISS medications, with the primary comparison focused on spaceflight-exposed samples relative to lot-matched terrestrial control samples. We hypothesized that spaceflight would impart detectable, albeit minor, changes in API stability compared with ground control samples.

Methods

Medication Preparation, Spaceflight, and Return to Earth

The Johnson Space Center (JSC) pharmacy purchased commercially available medications from pharmaceutical

wholesalers/distributors. Medications were purchased with a sufficient number of units to allow lot-matched medications to be randomized into 2 groups: spaceflight exposure and terrestrial controls, ensuring equivalent shelf life at the time of randomization. Spaceflight medications and the corresponding control samples were either repackaged into amber (light-protected) polymeric zipper-lock bags (Minigrip Reclosable Bags, Consolidated Plastics, Stow, OH) or kept in the manufacturer's sealed packaging per JSC pharmacy procedures for supplying medications to the ISS (see online [Supplementary Table S1](#) and online [Supplementary Figure S1](#)). These packaging approaches reflected routine NASA ISS HMS pharmacy logistics and were not selected or modified for this study to provide pharmaceutical-specific isolation from ambient carbon dioxide, humidity, or the spacecraft radiation environment. Terrestrial control samples were placed into an environmentally controlled chamber (Versatile Environmental Test Chamber MLR-350H, Sanyo Denki, Torrance, CA) and maintained at 21 to 23°C and between 35 and 49% relative humidity at JSC. Experimental medications were flown to the ISS and kept at standard ISS environmental conditions, which ranged from 18 to 27°C and between 25 and 75% relative humidity and which are within the United States Pharmacopoeia (USP)-defined allowable excursions for temperature and humidity.¹⁵ Terrestrial control conditions were selected to approximate USP-defined controlled room temperature storage (20–25°C, allowable excursions 15–30°C) with moderate humidity control, whereas ISS environmental conditions reflect actual operational spacecraft ranges rather than pharmaceutical-specific storage optimization. This study originated from an operational medication deployment and return process. As a result, baseline API testing was not designed into the original workflow. Subsequent pandemic-related programmatic disruption further limited the opportunity to modify the design before final analysis was completed by the present team.

Following typical JSC pharmacy operational protocols and scheduled launch opportunity, ISS medications undergo periodic resupply with crewmembers replacing older, unused medications from ISS HMS kits with newly supplied drug products.⁶ These medications were down massed by the next available return vehicle. This study analyzed a convenience sample of medications that were launched on SpaceX Commercial Resupply Service missions 15, 16, 17, 18, and 20 (SpX-15, -16, -17, -18, and -19) and Northrop Grumman's Cygnus NG-11 resupply mission ([Table 1](#)). Medications were not included on all resupply missions because ISS HMS pharmacy restocking is performed as needed based on inventory, utilization, and shelf-life considerations. The returned samples available for analysis did not follow a uniform study-wide resupply frequency or fixed pre-expiration removal threshold, and medications were included based on operational

availability. This convenience sample was selected from a larger pool of returned medications based on mission relevance, variety of liquid and solid formulations, attempts to maintain consistent manufacturers across flights, samples of the same medication across multiple flights, maximum time on the ISS, and amount of available research funding. At launch, all medications were within their labeled expiration period, with an average of 525 ± 200 d remaining until expiration across all missions (see online [Supplementary Table S1](#)).

Environmental conditions during launch, landing, and postlanding recovery were not directly measured for the study samples. However, these phases were conducted under standard NASA and commercial resupply vehicle operational constraints designed to maintain payload integrity. Following return and postflight processing by NASA, the spaceflight-exposed medications were transferred to the environmentally controlled chambers at JSC and stored along with the corresponding ground control samples until postflight analytic testing could be performed. During this interval, which was prolonged by disruption to the original analytic timeline during the COVID-19 pandemic, all pharmaceuticals remained inside the same JSC pharmacy or manufacturer packaging and were maintained together in the same environmentally controlled chambers. After transfer to the Lackland Air Force Base Clinical Investigations and Research Support Laboratory in March 2023, the medications were stored in a temperature-controlled room ($23\text{--}24^\circ\text{C}$, $35\text{--}50\%$ relative humidity) prior to testing. All analytic testing was completed within 6 mo of receipt of the medications. Although all medications were within their labeled expiration period at launch, all had expired by final content analysis after operational use, post-return storage, and delayed laboratory testing.

Reagents and Standards Preparation

Caffeine, diazepam, diphenhydramine, ketamine, lidocaine, and naloxone certified reference standards were obtained from Cerilliant Corporation (Round Rock, TX) as solutions at $1\text{ mg}\cdot\text{mL}^{-1}$ in methanol (MeOH). Epinephrine (solid) and promethazine ($1\text{ mg}\cdot\text{mL}^{-1}$ in MeOH containing 0.1% hydrochloric acid) reference standards were obtained from Cayman Chemicals (Ann Arbor, MI). High-performance liquid chromatography (HPLC) grade ($>99.9\%$ purity; Fisher Scientific, Pittsburgh, PA) acetonitrile, MeOH, water, and formic acid solvents were used as diluents and mobile phases. All calibrator solutions were prepared in 10% MeOH (Diluent 1) except for promethazine, which was prepared in 10% MeOH + 0.1% formic acid (Diluent 2). Three nonzero calibrators (25, 50, and 100%) containing 7 of 8 analytes were prepared in Diluent 1 according to the determined target concentration of each analyte (see online [Supplementary](#)

[Table S2](#)). A separate set of 3 nonzero promethazine calibrators (25, 50, and 100%) was prepared in Diluent 2. Calibrators were prepared serially with the appropriate diluent, and a quality control solution at 75% calibrant was prepared by mixing equal volumes of the 100 and 50% calibrators. All calibrators were stored as 1.2-mL aliquots at $5 \pm 3^\circ\text{C}$ for immediate use (within 1 wk) or at $-20 \pm 3^\circ\text{C}$ for long-term storage. Calibrator aliquots were sampled repeatedly throughout analysis until exhausted. Additional information regarding method validation, such as linearity, limits of quantification, carryover studies, and precision/accuracy, is available in the online Supplementary Methods file.

Sample Preparation

Method development and sample testing were performed by a single chemist (DL). Testing prioritized measured absolute API potency rather than USP methods for full pharmaceutical quality assessment. Matched spaceflight and ground control samples from the same lot were analyzed together in a single calibrated batch, with 10 replicates tested for each medication. These replicates were distributed across multiple vials, tablets, or capsules for each medication when feasible. When limited numbers of liquid medications were available, individual vials were subdivided to yield 10 replicate analyses. Samplings were diluted in the appropriate diluent (1 or 2) to final concentration (dilution factors are listed in online [Supplementary Table S2](#)). The final aliquot of each medication was transferred to a clean, labeled high-performance liquid chromatography vial. When applicable, different medications were pooled into the same vial and analyzed concurrently to minimize dilution and total run time. Additional details on replicate preparation are provided in the online Supplementary Methods file.

Instrumentation

Chromatography was performed using a Waters ACQUITY UPLC System (Waters Corp, Milford, MA) consisting of a temperature-controlled sample manager, binary pump, and temperature-controlled column compartment. Chromatographic separation was performed on a Waters ACQUITY UPLC BEH C18 (2.1×100 mm, $1.7\text{-}\mu\text{m}$) column with temperature regulated at 35°C . Mass spectrometry was performed in tandem with ultra-high-pressure liquid chromatography (UPLC) using a Waters ACQUITY Tandem Quadrupole Detector (Waters Corp) operated in positive electrospray ionization mode with multiple reaction monitoring using MassLynx Mass Spectrometry Software (Waters Corp). Additional instrumentation details are available in the online Supplementary Methods file, and the individual cone voltages, collision energies, and other multiple reaction

Table 1. Medication Characteristics by Spaceflight Mission.

Medication	Formulation	Dosage	Days in space	Days expired	Medication	Formulation	Dosage	Days in space	Days expired
<i>SpX-15, launch: June 29, 2018</i>									
Caffeine	Tablet	200 mg	199	1584	SpX-18, launch: July 25, 2019	Tablet	200 mg	166	670
Diphenhydramine	Capsule	25 mg	424	1096	Caffeine	Solution	5 mg·mL ⁻¹	166	1278
	Solution	50 mg·mL ⁻¹	424	1523	Diphenhydramine	Capsule	25 mg	257	731
Epinephrine	Autoinjector	0.1 mg·mL ⁻¹	424	1614		Solution	50 mg·mL ⁻¹	257	1065
Lidocaine	Solution	10 mg·mL ⁻¹	424	1523	Ketamine	Solution	50 mg·mL ⁻¹	257	823
Naloxone	Solution	1 mg·mL ⁻¹	424	1462	Lidocaine	Solution	10 mg·mL ⁻¹	257	670
Promethazine	Solution	25 mg·mL ⁻¹	424	1554	Naloxone	Solution	1 mg·mL ⁻¹	257	1096
	Tablet	25 mg	199	1676	Promethazine	Solution	25 mg·mL ⁻¹	257	1096
<i>SpX-16, launch: December 5, 2018</i>									
Caffeine	Tablet	200 mg	265	853	SpX-20, launch: March 7, 2020	Tablet	25 mg	257	1157
Promethazine	Tablet	25 mg	265	1310	Caffeine	Tablet	200 mg	313	549
<i>SpX-17, launch: May 4, 2019</i>									
Epinephrine	Autoinjector	0.1 mg·mL ⁻¹	248	1338	Diazepam	Solution	5 mg·mL ⁻¹	972	930
					Diphenhydramine	Capsule	25 mg	490	580
						Solution	50 mg·mL ⁻¹	490	823
					Ketamine	Solution	50 mg·mL ⁻¹	490	670
					Lidocaine	Solution	10 mg·mL ⁻¹	490	670
					Naloxone	Solution	1 mg·mL ⁻¹	573	731
					Promethazine	Solution	25 mg·mL ⁻¹	490	792
<i>NG-11, launch: April 17, 2019</i>									
Diphenhydramine	Solution	50 mg·mL ⁻¹	132	1065		Tablet	25 mg	313	1065
Lidocaine	Solution	10 mg·mL ⁻¹	132	945					
Promethazine	Solution	25 mg·mL ⁻¹	132	1096					

Note. Medications included in the analysis are listed by formulation, dosage, manufacturer, expiration timing, days in space aboard the International Space Station, and days expired at the start of chemical analysis. Missions are displayed from earliest to latest, left to right. Days in space were calculated from launch to landing, and days expired were calculated from the manufacturer expiration date to the initiation of postflight analytic testing.

monitoring parameters for each medication are described in online [Supplementary Table S3](#). All data were processed using TargetLynx software. During postprocessing, dilution factors and correction adjustments were applied to account for differences in chemical forms between the standard and the medication. The API potency, expressed as a percent of label, was then calculated by comparing the measured dosage with the manufacturer's stated dosage.

Statistical Analysis

To quantify differences between spaceflight-exposed medications and lot-matched ground control samples, percent difference in API content was calculated for paired samples and normalized to the corresponding control value $[(\text{spaceflight exposed} - \text{control})/\text{control} \times 100]$. An a priori threshold of a 5% difference was selected as a conservative criterion for defining meaningful deviations between groups, informed by prior pharmaceutical stability literature and chosen to be more stringent than the commonly applied 10% thresholds.⁸ Absolute API content was evaluated relative to 2 a priori ranges: a narrow 95 to 105% of labeled dosage and a broader 80 to 120% range.¹⁶ The 80 to 120% range was arbitrarily selected to allow broader descriptive interpretation of measured API content across medications and formulations. These ranges were not intended to define clinical acceptability, therapeutic equivalence, regulatory acceptability, or medication safety. Secondary analyses evaluated associations between API potency and days expired (defined as days from manufacturer expiration date to the date of initiation of post-flight analytic testing) as well as between potency and days in space (defined as days from launch to landing). These analyses were conducted for both spaceflight-exposed and control samples to distinguish natural degradation trends from effects potentially attributable to the spaceflight environment.

Because this study analyzed an operational convenience sample of returned medications, no prospective power calculation was performed. Instead, available lot-matched samples were analyzed, and replicate testing was used to maximize precision within the available sample set. GraphPad Prism (GraphPad Software, Boston, MA) was used for statistical analyses. Two-way analyses of variance (ANOVA) was used to assess the difference among lot-matched ground control samples and spaceflight-exposed medication potency for each flight with Tukey's HSD (honestly significant difference) post-hoc testing. Mixed-effects analyses were employed when replicate potency data from either the ground control samples or spaceflight-exposed medications were limited. Linear regression analysis was used to analyze the effect of days expired and days in space on medication potency. Given that some medications appeared on only 2 flights, days in space could not be correlated against days expired

because the linear regression is deterministic. A 2-sided alpha level of $P=0.05$ was set for all statistical tests.

Results

Experimental medications were exposed to spaceflight for 132 to 972 d, varying by medication and mission ([Table 1](#)). Across all medications and flights, the average number of days in space was 339 ± 171 d. Diphenhydramine (solution), lidocaine, and promethazine (solution) flown on the Cygnus NG-11 mission had the shortest spaceflight exposure of 132 d, whereas diazepam flown on SpX-20 mission had the longest exposure of 972 d. The manufacturer of each medication was the same across flights except for caffeine, diphenhydramine (solution), and promethazine (solution and tablet) (see online [Supplementary Table S1](#)). All pharmaceuticals were expired at the time of content analysis, with an average days expired across all medications of 1063 ± 339 d (see [Table 1](#)). The caffeine tablets flown on the SpX-20 mission had the shortest days expired (549 d), whereas the promethazine tablet flown on SpX-15 mission had the longest days expired (1676 d).

[Figure 1](#) shows the percent difference in API content between spaceflight-exposed and lot-matched ground control medications averaged across all available flights. Percent differences clustered primarily within $\pm 5\%$, indicating close API agreement between spaceflight-exposed and ground control samples despite wide variation in days in space and days expired. Epinephrine demonstrated the largest negative percent difference (-10%), reflecting lower API content in flown samples relative to control samples. Caffeine tablets and diphenhydramine solution also exhibited average percent differences that exceeded the $\pm 5\%$ threshold at -6% and $+8\%$, respectively.

Absolute API potency was evaluated relative to 2 a priori ranges, 95 to 105% and 80 to 120% of labeled dosage for solid ([Figure 2](#)) and nonsolid formulations ([Figure 3](#)). For solid formulations evaluated in this study, 8 of 11 spaceflight-exposed samples (73%) and 6 of 11 ground control samples (55%) fell outside the 95 to 105% range. No spaceflight-exposed or ground control solid-formulation measurements fell outside the broader 80 to 120% range on any flight. Two-way ANOVA identified a statistically significant exposure-related difference for promethazine tablets ($P=0.0001$), with post-hoc testing localizing this difference to the SpX-15, SpX-16, and SpX-18 missions. However, the absolute separation between ground and spaceflight promethazine tablets remained less than $\pm 5\%$.

Similarly, several nonsolid medications fell outside the 95 to 105% range in both ground control samples (17 of 21; 81%) and spaceflight-exposed samples (19 of 21; 90%) ([Figure 3](#)). Epinephrine, ketamine, and promethazine (solution) fell below the 80% API threshold on at least

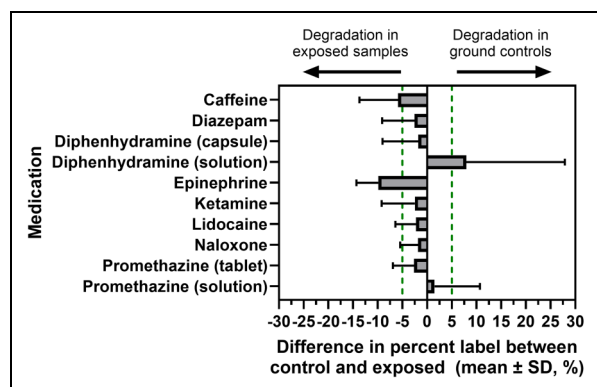


Figure 1. Percent difference in active pharmaceutical ingredients between ground and spaceflight medications. *Note.* Averaged percent difference calculated as (spaceflight exposed – control)/control $\times$ 100 across all possible flights. Negative values indicate lower active pharmaceutical ingredient content in spaceflight-exposed samples relative to control samples, whereas positive values indicate higher active pharmaceutical ingredient content in exposed samples. Dashed green lines indicate 5% a priori cutoff between control and spaceflight-exposed medications.

1 flight, with epinephrine exhibiting the most pronounced reductions in API content relative to control samples. Two-way ANOVA detected statistically significant exposure-related differences for diphenhydramine solution ($P=0.0286$), epinephrine ($P<0.0001$), lidocaine ($P=0.0125$), and naloxone ($P=0.0080$). Except for epinephrine, the between-group separation was less than $\pm 5\%$ and generally remained within the broader 80 to 120% range. Of note, the spaceflight-exposed epinephrine was, on average, 10% less potent than ground control samples (see Figure 3). These changes corresponded with visual color differences observed in both flight and ground

control epinephrine (see online [Supplementary Figure S2](#)), suggesting the formation of adrenochrome due to the oxidation and cyclization of epinephrine.¹⁷

API potency for both the ground control sample and spaceflight-exposed medications was correlated against days in space (Figure 4) and days expired (Figure 5). Among spaceflight-exposed medications, days in space was negatively correlated with potency for both formulations of promethazine. In contrast, days in space was positively correlated with potency for spaceflight-exposed caffeine, diazepam, epinephrine, and ketamine. When regression slopes for spaceflight-exposed (days in space vs API) and lot-matched control (days in storage vs API) samples were compared, no statistically significant between-group differences were identified. API potency changes were negatively correlated with days expired for 50% (5 of 10) of the medications (ie, diazepam, diphenhydramine [capsules], ketamine, lidocaine, and naloxone), whereas API potency changes were positively correlated with days expired for 30% (3 of 10) of medications (ie, diphenhydramine [solution], epinephrine, and promethazine [solution, ground control tablet]). There were no significant differences between spaceflight-exposed and ground control sample regression lines assessing days expired vs API.

Discussion

In this convenience sample of operational medications flown aboard the ISS, API potency was generally similar between spaceflight-exposed samples and lot-matched ground control samples. Most between-group differences were within $\pm 5\%$ relative to controls in the setting of prolonged spaceflight exposure and delayed postflight analysis. Epinephrine represented the clearest exception, with

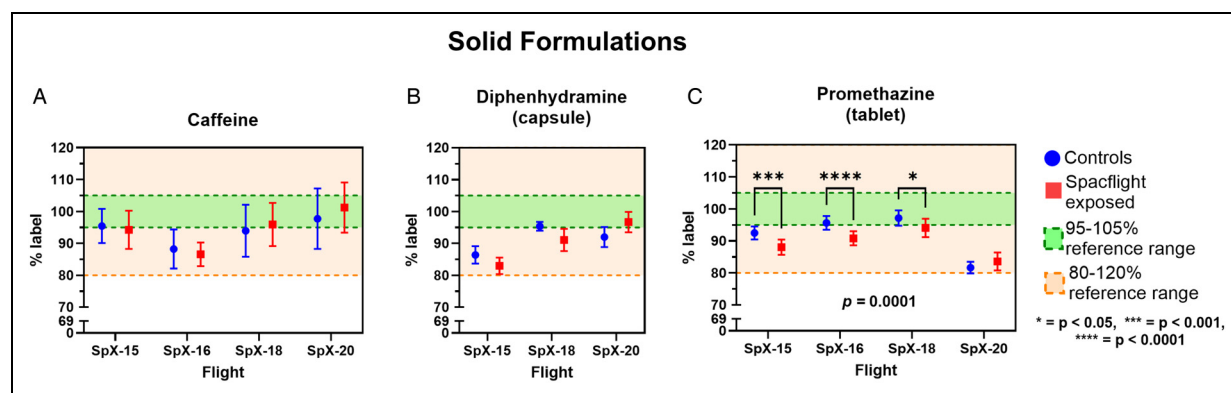


Figure 2. Comparison of active pharmaceutical ingredient content (% label) for ground vs spaceflight-exposed solid formulations. *Note.* Absolute active pharmaceutical ingredient content expressed as percent of label (mean $\pm$ SD) between ground control (blue, circle) and spaceflight-exposed (red, square) medications for (A) caffeine, (B) diphenhydramine (25-mg capsule), and (C) promethazine (25-mg tablet). Significance determined by 2-way analysis of variance; main effect P values displayed for medications with significant differences between exposure and post-hoc analyses for each flight are indicated as * $P<0.05$, *** $P<0.001$, and **** $P<0.0001$. The a priori ranges of 95 to 105% and 80 to 120% are overlaid in green and orange, respectively.

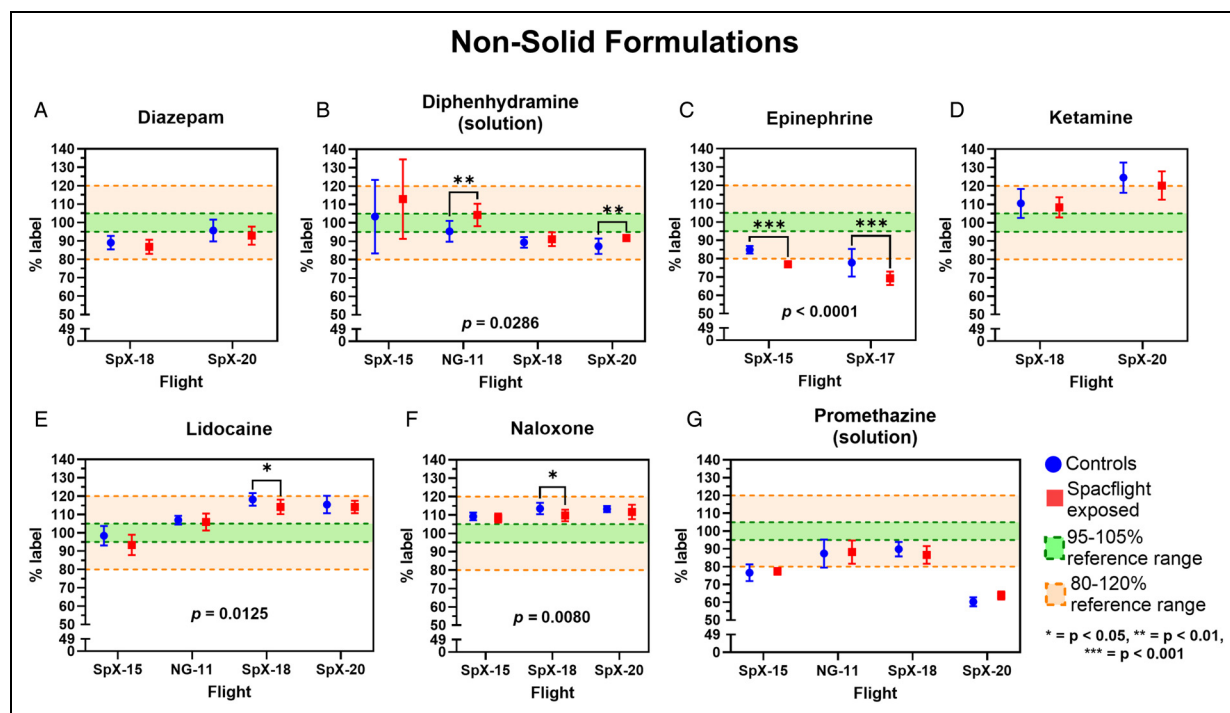


Figure 3. Comparison of active pharmaceutical ingredient content (% label) for ground versus spaceflight nonsolid formulations. Note. Absolute active pharmaceutical ingredient content expressed as percent of label (mean $\pm$ SD) between ground control (blue, circle) and spaceflight-exposed (red, square) medications for (A) diazepam, (B) diphenhydramine (50 mg·mL⁻¹ solution), (C) epinephrine, (D) ketamine, (E) lidocaine, (F) naloxone, and (G) promethazine (25 mg·mL⁻¹ solution). Significance determined by 2-way analysis of variance; main effect P values displayed for medications with significant differences between exposure and post-hoc analyses for each flight are indicated as * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$. The a priori ranges of 95 to 105% and 80 to 120% are overlaid in green and orange, respectively.

lower potency in spaceflight-exposed samples than in control samples. Within the limitations of the study design, these results suggest that ISS spaceflight exposure was associated with limited relative potency changes for most medications in this sample.

Medication degradation during spaceflight may be influenced by several factors, including continuous galactic cosmic radiation exposure, API chemical structure, inactive ingredients, packaging, and duration of exposure.^{5,6,12} Prior studies have evaluated a relatively small number of medications, and most have not included comparisons with lot-matched ground control samples.^{8–10,18} These studies often have focused on whether medications crossed predefined potency thresholds while providing less context for interpreting modest potency changes.⁸ Our analysis builds on this prior work by evaluating operationally deployed medications across multiple formulations and ISS missions with lot-matched ground control samples as the primary comparator. To focus the between-group comparison, we used a prespecified $\pm 5\%$ threshold to describe the magnitude of paired differences between spaceflight-exposed and control samples. We then interpreted absolute API potency using 2 prespecified ranges, 95 to 105% and 80 to 120% of labeled dosage.

Because the medications varied in flight duration, storage duration, formulation, and time past expiration, no single threshold fully captures the operational relevance of the results. We used complementary interpretive frameworks to contextualize the observed potency changes. The 95 to 105% range provides a stringent drug quality reference, the 80 to 120% range provides broader context for absolute API potency, and the paired $\pm 5\%$ threshold helps describe whether spaceflight-exposed samples diverged from their lot-matched control samples.

Because the medications varied in both days in space and days expired, we used regression analyses to examine whether these time variables were associated with API potency. These analyses identified medication-specific associations with both days in space and days expired, with the direction of these associations varying by medication. This variation is important because positive associations with time do not necessarily indicate improved stability, particularly when the same medications also demonstrated reduced absolute API potency. Higher measured API in some samples may reflect formulation, excipient, or sample preparation effects that alter apparent relative concentrations. Therefore, the regression analyses are best interpreted as exploratory rather than evidence that

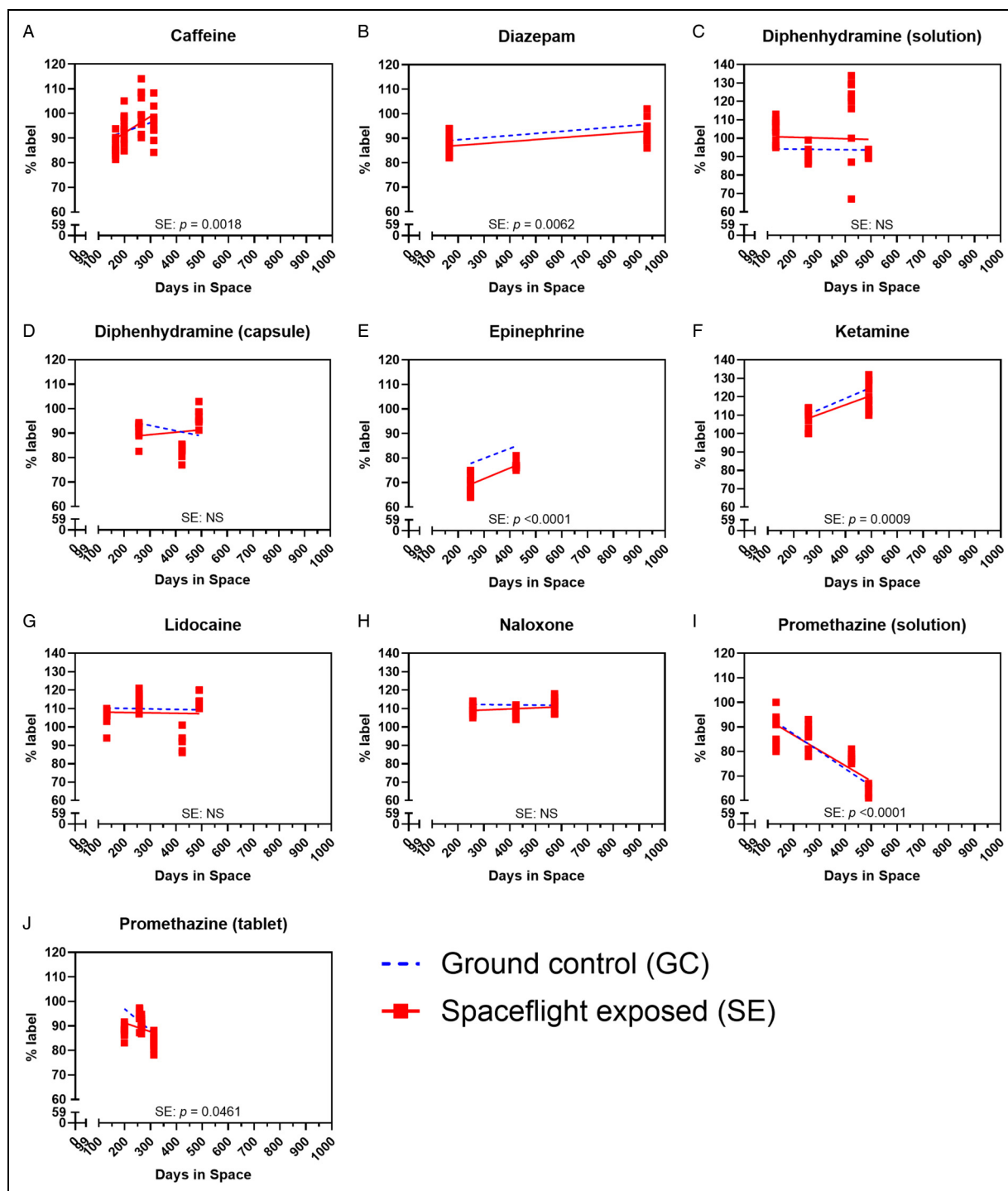


Figure 4. Effect of days in space on medication potency. *Note.* Correlation between days in space and active pharmaceutical ingredient potency expressed as percent label (% label) for spaceflight-exposed samples (red squares), where each data point represents an individual spaceflight, for (A) caffeine, (B) diazepam, (C) diphenhydramine (solution), (D) diphenhydramine (capsule), (E) epinephrine, (F) ketamine, (G) lidocaine, (H) naloxone, (I) promethazine (solution), and (J) promethazine (tablet). The observed linear regression of spaceflight-exposed samples is displayed as red solid lines. The linear regression of lot-matched ground control samples during terrestrial storage is displayed as blue dashed lines for reference. However, because these control medications were not exposed to spaceflight, the individual lot-matched control results are not visually displayed. Significant positive or negative correlations are overlaid. NS indicates nonsignificance.

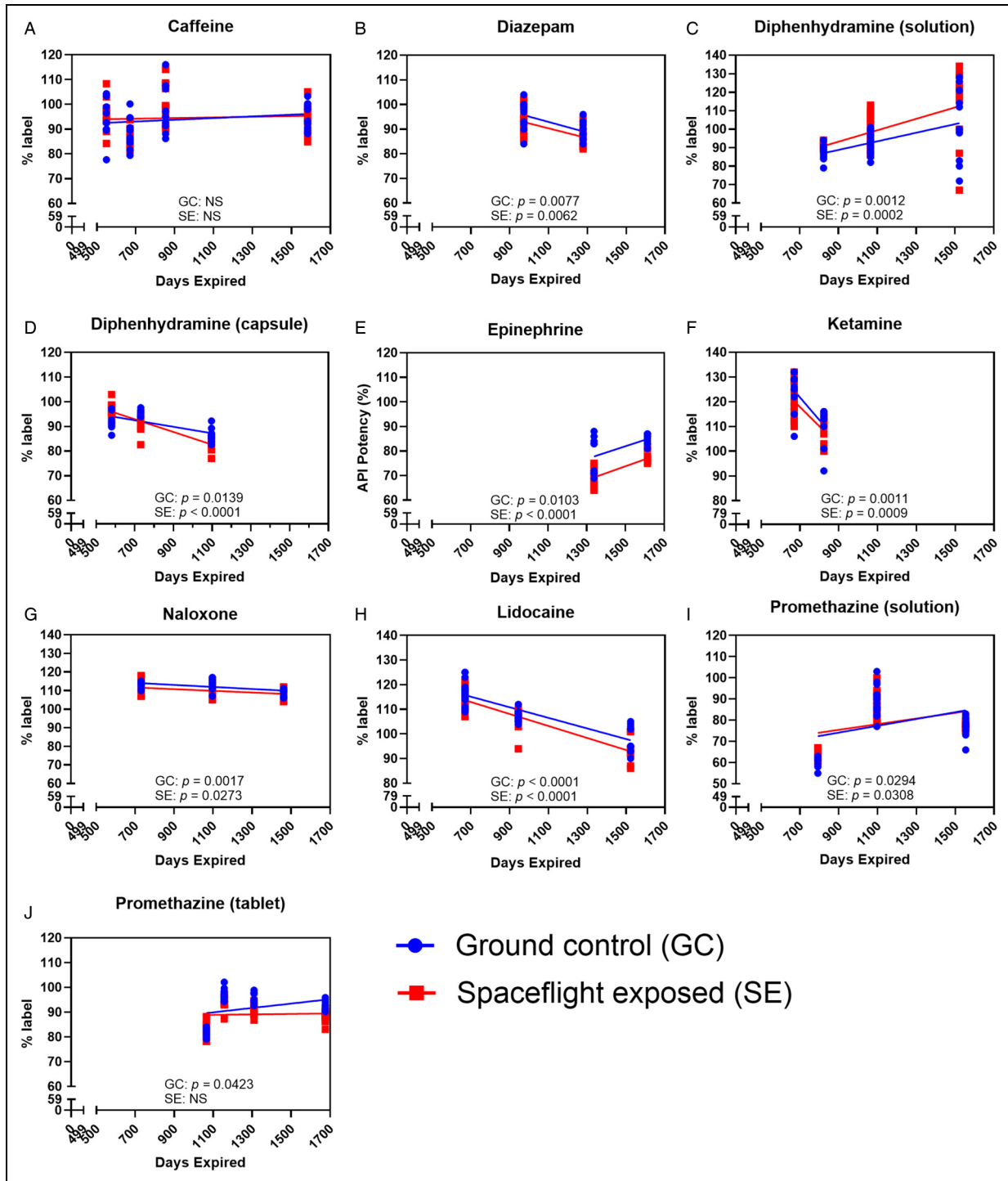


Figure 5. Effect of days expired on medication potency. *Note.* Correlation between days expired and active pharmaceutical ingredient potency expressed as percent label (% label) for (A) caffeine, (B) diazepam, (C) diphenhydramine (solution), (D) diphenhydramine (capsule), (E) epinephrine, (F) ketamine, (G) naloxone, (H) lidocaine, (I) promethazine (solution), and (J) promethazine (tablet) from ground control (blue circles) and spaceflight-exposed samples (red squares), where each data point represents an individual lot-matched sample from a single spaceflight or its corresponding ground control. Linear regression of ground control samples is shown as blue solid lines, and linear regression of spaceflight-exposed samples is shown as red solid lines. Because days expired applies equally to ground control and spaceflight-exposed medications, individual data points for both groups are included. Significant positive or negative correlations are overlaid. NS indicates nonsignificance.

either days in space or days expired independently drove potency changes. The absence of baseline preflight measurements and the prolonged postflight storage interval further limit our ability to assign these temporal changes to any single factor.

Epinephrine was the clearest medication-specific finding in this study. It demonstrated lower potency in spaceflight-exposed samples relative to control samples and was the only medication with both consistent between-group divergence and absolute potency below the lower bound of the broader 80 to 120% range. This finding is consistent with the known instability of aqueous epinephrine formulations,^{19,20} which are susceptible to oxidative degradation. Other medications, including promethazine, showed reductions in API potency in both spaceflight-exposed and terrestrial control samples, suggesting that storage duration, formulation stability, and sample-specific factors likely contributed to some observed potency changes. For long-duration missions, medications with greater storage-related vulnerability, particularly epinephrine, may require medication-specific mitigation strategies, such as shorter replacement intervals or evaluation of alternative formulations.

The medications in this study experienced medium- to long-duration ISS exposures followed by prolonged terrestrial storage. Although the prolonged terrestrial storage complicates attribution of potency changes solely to spaceflight exposure, this scenario approximates some aspects of delayed-use medical logistics for lunar or Martian missions.²¹ The ISS remains a relatively controlled environment compared with several potential exploration mission stressors. Medications used during future missions may experience depressurization or vacuum exposure, higher cumulative radiation environments beyond low-Earth orbit, longer storage intervals, and more variable temperature or humidity conditions.^{22,23} Prior analysis has suggested that spaceflight may modestly accelerate natural degradation processes, supporting an operational approach focused on medication-specific vulnerability rather than broad formulary-wide assumptions.⁵ Mission planners may need to account for storage duration, formulation, packaging, replacement intervals, and expected clinical use, particularly for medications with known instability or supplies that are prepositioned for delayed use. Modest API potency changes also may carry greater operational significance if spaceflight-associated changes in pharmacokinetics or pharmacodynamics alter drug exposure, therapeutic response, or adverse-effect profiles.²⁴

These conclusions must take into account several limitations. The primary limitation is that this study represents a post-storage comparison of operational medications rather than a prospective longitudinal stability study. All medications were expired at the time of analysis, baseline preflight API content was not measured, and the

pandemic-related delay prolonged the interval between return and final testing. As a result, these findings should be interpreted as comparisons between lot-matched post-storage samples rather than measured changes from preflight baseline. Second, variations in duration of spaceflight exposure existed both within individual medication types and between flights, and medications were not evenly represented across missions. These differences in days in space, days expired, formulation, and manufacturer limited the ability to perform more advanced statistical modeling or isolate the effect of any single exposure variable. Third, the limited sample size restricted our ability to draw definitive conclusions on the impact of repackaging on medication stability in spaceflight conditions. Several medications (eg, caffeine, diphenhydramine [solution], and promethazine [tablet and solution]) were sourced from multiple manufacturers. We cannot determine how differences in manufacturing processes may have influenced long-duration stability or measured API content. In addition, no procedures were performed during sample preparation to remove potential excipients from the medications, which may have contributed to variability in measured API content. Fourth, environmental conditions during launch, landing, and postlanding recovery were not directly measured for the study samples. Lastly, medication analysis required manual weighing and dilution, which introduces the potential for minor variability due to the high fidelity of the UPLC-MS instrumentation. We attempted to limit the intra-medication and intra-flight effects by having a single chemist perform the preparation and analysis of each medication. Future studies should aim to control for confounding variables, including manufacturer differences, spaceflight duration, and postexpiration storage time to improve the reliability of pharmaceutical stability assessments in spaceflight conditions.

Conclusion

In this lot-matched post-storage comparison of operational medications from the ISS HMS, most spaceflight-exposed samples demonstrated small relative differences in API potency compared with ground control samples. Epinephrine represented the clearest exception, with lower potency in spaceflight-exposed samples. These findings suggest that within the limitations of this study design, spaceflight exposure was not associated with large relative potency changes for most medications in our sample.

Author Note

An artificial intelligence-based language model (ChatGPT 5.5, OpenAI) was used as a writing support tool during manuscript preparation. The tool assisted with editing for clarity, tone, organization, and grammar and with refining phrasing in response to detailed author input. All study concepts, analyses,

interpretations, and conclusions were developed by the authors. ChatGPT 5.5 did not generate original scientific ideas, perform analyses, make independent interpretive decisions, or generate citations. All content was reviewed, edited, and approved by the authors, who take full responsibility for the manuscript.

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Lyle Babcock: Formal analysis; Writing – review & editing.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Prior Presentations

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Supplemental Material

Supplementary material for this article is available online.

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